

Introduction to Nutrition

Unit 1: Fundamentals of Nutrition

CHALLENGE 1.1 - PLANNING A HEALTH DIET

1.1 - 1. SOCIAL MEDIA, MIXED MESSAGES & MISINFORMATION

Q. In order for nutrition info to be considered trustworthy, it must be science-based, peer reviewed and what else? Replicable.

Q. You can generally trust nutritional info that comes from qualified professionals. What credential is given to a nutrition expert that must complete a specialized degree in nutrition, dietetics, public health or another related science? Registered Dietitian Nutritionist

1. INTRODUCTION TO NUTRITION

- Nutrition as a discipline is a young field - started in the 1930's - but its dynamic
- Nutrition can also be described as the sum of all processes involved in how organisms obtain nutrients, metabolize them, and use them to support all of life's processes.
- Trustworthy nutrition information is science-based, peer-reviewed, and replicable. Nutrition misinformation is not supported by science and may be misleading or incomplete.

1. WHO CAN YOU TRUST?

a. NUTRITIONAL EXPERTS

- Criteria are required to become an RDN:
 - Complete a minimum of a bachelor's degree
 - Complete a supervised practice program that is accredited (typically these are 6-12 month programs)
 - Pass an exam given by Commission on Dietetic Registration (CDR)
 - Continue to complete continuing education requirements each year

b. RELIABLE MEDIA SOURCES

- What is the most reliable source for food information that is regulated by the Food and Drug Administration (FDA)? The nutrition label
- Look at citation, sources, and qualifications.

3. INFORMED CONSUMER

- Does the advertisement claim Food and Drug Administration (FDA) approval? It is illegal to suggest FDA approval as a part of any marketing claim. However, all medical products sold across state lines must be registered with the FDA. Ask for the FDA proof of product listing if in doubt.

1.1 - 2. PERSONAL FOOD CHOICES

Q. What factors are “primary reasons” for selecting certain foods?
A. Taste + hunger

Q. Which term means “feeling full or satisfaction of appetite”?
A. Satiety

Q. In general, which taste do humans prefer?
A. Sweet

1. BIOLOGICAL FACTORS AFFECTING CHOOSING FOODS

- Satiety, Palatability, Texture
- Generally, foods higher in protein and carbohydrate are the most satiating while fat is the least satiating.

2. ENVIRONMENTAL FACTORS AFFECTING CHOOSING FOODS

- Weather, climate, setting, buffer layout, product layout, fast food vs sit down, availability

3. SOCIAL AND CULTURAL FACTORS AFFECTING CHOOSING FOODS

- Tradition, family setting, income, cultural intolerance / tolerance, religion rules

4. YOUR HEALTH STATUS AFFECTS CHOOSING FOODS

- *Hypertension*: May require a low-sodium diet.
- *Diabetes*: May need a consistent carbohydrate diet.
- *Women*: Women of childbearing age need a diet rich in iron and folic acid.
- *Pregnancy*: Pregnant women may experience food aversions, potentially impacting their nutritional intake.
- *Aging*: As people age, their taste preferences shift, often with an increased preference for sweetness.

1.1 - 3. BENEFITS OF HEALTHY EATING

Q. Why should we avoid highly processed foods as part of a healthy diet? Highly processed food add unneeded sodium, saturated fats, and sugar to many diet, increasing the risk of chronic diseases.

Q. Choosing foods that do not have ____ can improve your cholesterol
A. Trans-fat

1. WHAT IS A HEALTHY DIET?

- Nutrient dense foods.

2. HOW DIETS AFFECT OUR HEALTH

- *Dietary Habits*: Most Americans do not maintain a healthy diet, with poor eating habits often starting from infancy.
- *Sodium*: Many Americans consume insufficient fruits and vegetables while exceeding recommended sodium intake.

- *Obesity*: A significant portion of the US population, both children and adults, are obese, leading to increased healthcare costs and health risks.

a. HEART HEALTH

- *Heart disease*: is the leading cause of death for adults in the United States. Eating foods low in saturated fats and high in fiber, increasing access to low-sodium foods, and regular physical activity can help prevent high blood cholesterol and high blood pressure.
- *DASH Diet*: The DASH diet, which is low in saturated fat and dietary cholesterol and rich in potassium, magnesium, calcium, and fiber, can help improve heart health.
- *Limiting certain types of fats*: reducing trans-fat can improve your cholesterol and decrease the risk of stroke or heart attack.

b. DECREASED RISK OF CANCER

- *Diet and Cancer Risk*: An unhealthy diet can increase the risk of some cancers, particularly those related to being overweight and obese.
- *Antioxidants*: Consuming foods rich in antioxidants can lower cancer risk by protecting cells from damage caused by free radicals.
- *Antioxidant-Rich Foods*: Fruits, vegetables, nuts, and legumes are good sources of antioxidants like beta-carotene, lycopene, and vitamins A, C, and E.

c. WEIGHT MAINTENANCE

- *Food Choices for Health*: Foods like fruits, vegetables, and beans are lower in calories than processed foods.
- *Benefits of Healthy Food Choices*: Choosing these foods helps maintain a healthy weight and reduces the risk of chronic disease.

d. OVERALL WELL-BEING

- *Mental Health Benefits*: Foods rich in omega 3 fatty acids, vitamin b12, folate, and vitamin D are linked to reduced depression and anxiety.
- *Cognitive Function Improvement*: Diets high in fruits and vegetables are associated with enhanced brain function and memory.
- *Body Image Enhancement*: Healthy eating can contribute to a positive body image, leading to improved self-esteem.

3. WAYS TO SUPPORT HEALTHY EATING

- **Healthy Eating Promotion in Schools**: CDC's Healthy Schools program promotes good nutrition in schools by providing guidelines and resources for healthier school meals, snacks, and water access.
- **Workplace Wellness Initiatives**: CDC encourages employers to offer healthier food options in cafeterias, snack shops, and vending machines, and supports breastfeeding mothers in the workplace.
- **Improving Food Access in Underserved Communities**: CDC funds programs to improve food systems in low-income and underserved areas by supporting food hubs, local stores, farmers' markets, and bodegas.
- **Low-Income Assistance Programs in the US**: WIC and SNAP are two programs that provide nutrition education, food assistance, and financial support for low-income individuals and families.

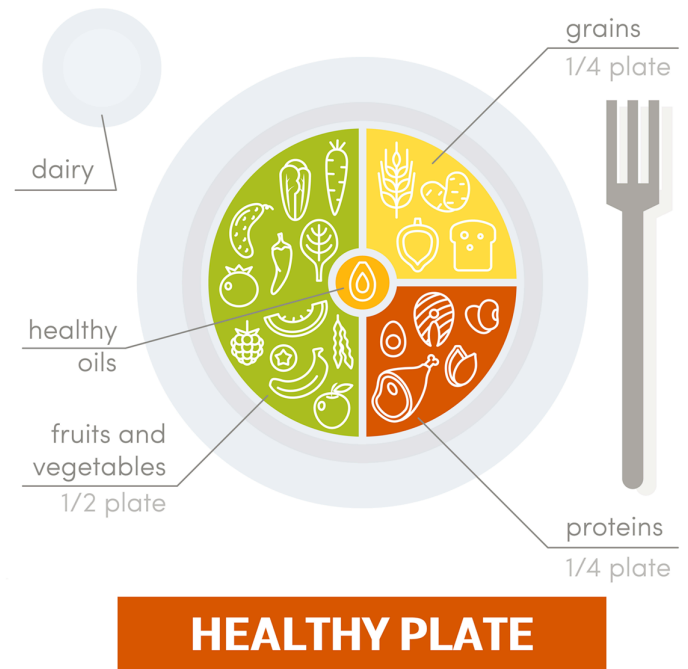
- **Healthy Eating Recommendations:** Swap sugary drinks for water, reduce red meat consumption, prioritize lean proteins, and fill half your plate with fruits and vegetables.
- **Individualized Nutrition:** Consulting a Registered Dietitian Nutritionist can help tailor dietary choices to individual needs based on factors like age, gender, and activity level.

1.1 - 4. FOUNDATIONS OF A HEALTHY DIET

Q. Which of the follow foods would be considered the most nutrient-dense?
Grapes

Q. Which of the follow is considered a macronutrient?
Carbohydrates

Q. Why are essential nutrients termed essential?
They must be supply by our diets because our bodies can't make them.



1. WHAT MAKES A HEALTHY DIET?

Healthy eating doesn't involve any particular diet. Rather, it means prioritizing your health by fueling your body with a variety of nutrient-rich foods.

a. NUTRIENTS

- **Essential Nutrients:** Nutrients that must be supplied by our diet because our bodies cannot make them, such as water.
- **Non-essential Nutrients:** Nutrients that our bodies can make, such as vitamin D.
- **Macronutrients and Micronutrients:** Macronutrients (carbohydrates, fat, and protein) are needed in large amounts, while micronutrients (vitamins and minerals) are needed in small amounts.

b. CALORIES

- The energy released from our macronutrients can be measured in calories.
- **Calorie Measurement:** Calories are small units of energy, and food energy is typically measured in kilocalories (kcalories, or uppercase Calories - 1000 calories).
- **Nutrient Density:** Nutrient-dense foods provide more beneficial nutrients relative to their calorie content.
- **Energy Density:** Energy-dense foods are high in calories but low in nutrients, often referred to as "empty calories."

2. CREATING BALANCE WHEN CHOOSING YOUR DIET

- **Adequacy:** Eating a variety of foods from all food groups ensures obtaining the right amounts of nutrients.
- **Eating:** Consuming foods in reasonable amounts, using portion control, and avoiding excessive intake of any single food.
- **Balance:** Prioritizing nutrient-dense foods over energy-dense foods to meet nutritional needs while allowing for occasional indulgence.

1.1 - 5. SCIENTIFIC APPROACH TO NUTRITION

Q. Study is conducted on if vitamin C makes people sleep longer. One group gets vitamin C, the other does not. The group that get vitamin C is known as?
Experiment Group

Q. Which of the following if tree of nutrition research?
New findings are constantly changing out recommendations for healthy living.

Q. Which of the following steps of the scientific method is when you would make an educated guess about what would happen in an experiment? Hypothesis

1. THE SCIENCE OF NUTRITION

- Nutrition is the science that links food to health and includes how humans ingest, digest, absorb, transport, use, and excrete food.
- Nutrition researchers use the scientific method to guide their work and help us make better choices for our health.

2. THE SCIENTIFIC METHOD

- Steps
 1. Propose a scientific or research question that will be interesting to investigate.
 2. Conduct research to examine other experiments that might impact the question.
 3. Formulate a hypothesis or an educated guess based on the research question.
 4. Design and conduct the experiment. A good design should take into account what has been done previously. Thus, a thorough review of methods and results published previously should be undertaken. This will help prevent making the same mistakes and save a lot of time performing the research.
 5. Analyze the the data and draw conclusions.
 6. If the hypothesis is not supported, then a new hypothesis/research question should be created and an experiment conducted accordingly.
 7. Communicate results (publish in peer reviewed journals).
- Experimental and Control Groups: To compare results and determine the effect of the treatment.
- Importance of Controlling Variables: To obtain the most accurate results by minimizing the influence of factors like athlete conditioning, age, or caffeine dosage.

3. THE DIFFERENT TYPES OF STUDIES

- Many nutrition studies are *observational*, focusing on the occurrence, distribution, and contributing factors of health problems in populations.
- *Epidemiology*: the study of the occurrence, distribution, and factors that contribute to health problems in populations.
- NHANES (National Health and Nutrition Examination Survey) and the Framingham Heart Study are examples of large-scale observational studies.
- *Limitations of Observational Studies*: While valuable for identifying relationships, observational studies cannot establish cause and effect.
- Scientific Method in Nutrition Science: Illustrated by the case of vitamin C and scurvy, where controlled trials and systematic reviews led to the discovery of the role of vitamin C in preventing scurvy.
- *Interpreting Research Findings*: Requires analyzing limitations, considering multiple high-quality studies, and relying on peer-reviewed research for evidence-based recommendations.
- Evidence-Based Practice in Nutrition: Nutrition and dietetics are evidence-based, meaning recommendations are based on the scientific method and constantly evolving with new research findings.

VOCAB

- *Nutrition*: the sum of all processes involved in how organisms obtain nutrients, metabolize them, and use them to support all of life's processes.
- *Nutrients*: A substance in food that can provide energy, contribute to body structure, and/or regulate body processes.
- *Health Claim*: Claims that have been reviewed by the FDA and are allowed on food products showing that a food or food component may reduce the risk of a disease or a health-related condition.
- *Health Fraud*: Fraud involving the selling of drugs, devices, foods, or cosmetics that have not been proven effective.
- *Nutrition Misinformation*: Information that is not supported by science and may be misleading or incomplete.
- *Registered Dietitian Nutritionist (RDN)*: A nutrition expert required to complete a specialized degree in dietetics, nutrition, public health, or other related sciences.
- *Diabetes*: A chronic (long-lasting) health condition that affects how your body turns food into energy.
- *Food Aversions*: A strong dislike of a specific food or foods.
- *Health Status*: Refers to your medical conditions (both physical and mental health), claims experience, receipt of health care, medical history, genetic information, evidence of insurability, and disability.
- *Hypertension*: Another name for high blood pressure. It can lead to severe health complications and increase the risk of heart disease, stroke, and sometimes death.

Blood pressure is the force that a person's blood exerts against the walls of their blood vessels.

- *Palatability*: Acceptable to the taste; sufficiently agreeable in flavor to be eaten.
- *Satiety*: The quality or state of being fed or gratified to or beyond capacity.
- *Seasonality*: Produce that is purchased and consumed around the time that it is harvested.
- *Antioxidants*: Substances that can prevent or slow damage to cells caused by free radicals, unstable molecules produced by the body in response to environmental and other pressures.
- *Cardiovascular Disease (CVD)*: A general term for conditions affecting the heart or blood vessels, typically associated with fatty deposits inside the arteries (atherosclerosis) and an increased risk of blood clots.
- *DASH Diet*: A flexible and balanced eating plan (Dietary Approaches to Stop Hypertension) designed to promote a heart-healthy eating style for life.
- *Nutrient-Dense Food*: Foods rich in vitamins, minerals, and other nutrients important for health, with minimal saturated fat, added sugars, and sodium.
- *Adequacy*: The sufficient intake of essential nutrients needed to fulfill nutritional requirements for optimal health.
- *Balance*: Consuming foods from multiple food groups to maintain optimal nutritional status.
- *Calorie*: A unit of energy obtained from the food and beverages consumed.
- *Energy-Dense Food*: Foods high in calories relative to the nutrients they contain.
- *Essential Nutrient*: Nutrients vital for health, including vitamins, minerals, proteins, fats, water, and carbohydrates.
- *Macronutrient*: Nutrients required in large amounts, such as carbohydrates, fats, and proteins.
- *Micronutrient*: Vitamins and minerals essential for healthy development, disease prevention, and overall wellbeing.
- *Moderation*: The practice of avoiding excess or extremes in consumption.
- *Epidemiology*: The study of the occurrence, distribution, and factors that contribute to health problems in populations.
- *National Health and Nutrition Examination Survey (NHANES)*: A program of studies designed to assess the health and nutritional status of adults and children in the United States.
- *Framingham Heart Study*: A long-term, ongoing cardiovascular cohort study of residents of the city of Framingham, Massachusetts. Much of the knowledge concerning heart disease, such as the effects of diet, exercise, and common medications such as aspirin, is based on this longitudinal study.

- *Scurvy*: A disease caused by a deficiency of vitamin C, characterized by swollen bleeding gums and the opening of previously healed wounds, which particularly affected poorly nourished sailors until the end of the 18th century.
- *Evidence-Based*: A practice that has been rigorously evaluated in experimental evaluations—like randomized controlled trials—and shown to make a positive, statistically significant difference in important outcomes.

CHALLENGE 1.2 - TOOLS FOR PLANNING A HEALTH DIET

1.2 - 1. FOOD PATTERNS AND DIETARY GUIDELINES

Q. Which breakfast would fall into the Healthy Vegetarian Pattern?
A. Raisin Bran, milk, coffee, low-fat milk, whole grain toast & scrambled egg.

1. THE DIETARY GUIDELINES

- *Food Group Plans*: Help build a balanced diet by selecting foods from different groups, each representing a unique set of nutrients.
- *USDA Dietary Guidelines*: Provide evidence-based recommendations for healthy eating patterns to promote health and prevent disease.
- *2020-2025 DGA Focus*: “Make Every Bite Count” with four overarching guidelines: follow a healthy dietary pattern, customize nutrient-dense choices, focus on food group needs, and stay within calorie limits.
- *Dietary Recommendations*: Limit added sugars, saturated fat, sodium, and alcoholic beverages.
- *Limits*: Less than 10% of calories from added sugars and saturated fat for ages 2+, less than 2,300mg sodium per day, and limit alcohol to 2 drinks/day for men and 1 drink/day for women.
- *Overall Eating Pattern*: Prioritize nutrient-rich foods, with limited room for added sugars, saturated fats, and alcohol.
- The Dietary Guidelines framework purposely provides recommendations by food groups and subgroups - not specific foods and beverages - to avoid being prescriptive. This framework approach ensures that people can “make it their own” by selecting healthy foods, beverages, meals, and snacks specific to their needs and preferences.

2. THE THREE USDA FOOD PATTERNS

- *Dietary Guidelines Recommendations*: Provide guidance on healthy eating across the lifespan.
- *USDA Food Patterns*: Identify daily amounts of foods from five major food groups and their subgroups to help individuals carry out Dietary Guidelines recommendations.

- *Three USDA Food Patterns:* Healthy U.S.-Style Pattern, Healthy Vegetarian Pattern, and Healthy Mediterranean-Style Pattern.

3. TOOLS FOR DEVELOPING DIETARY PATTERNS

- *Balanced Diet:* All food groups are essential, and daily selections should be made from each, with amounts varying based on individual energy needs.
- *USDA Food Patterns:* Recommended amounts and limits are set at 12 calorie levels, ranging from 1,000 to 3,200 calories, to meet the nutritional needs of individuals of different ages and activity levels.

Daily Amount of Food From Each Group												
Calorie Level ¹	1,000	1,200	1,400	1,600	1,800	2,000	2,200	2,400	2,600	2,800	3,000	3,200
Fruits ²	1 cup	1 cup	1½ cups	1½ cups	1½ cups	2 cups	2 cups	2 cups	2 cups	2½ cups	2½ cups	2½ cups
Vegetables ³	1 cup	1½ cups	1½ cups	2 cups	2½ cups	2½ cups	3 cups	3 cups	3½ cups	3½ cups	4 cups	4 cups
Grains ⁴	3 oz-eq	4 oz-eq	5 oz-eq	5 oz-eq	6 oz-eq	6 oz-eq	7 oz-eq	8 oz-eq	9 oz-eq	10 oz-eq	10 oz-eq	10 oz-eq
Protein Foods ⁵	2 oz-eq	3 oz-eq	4 oz-eq	5 oz-eq	5 oz-eq	5½ oz-eq	6 oz-eq	6½ oz-eq	6½ oz-eq	7 oz-eq	7 oz-eq	7 oz-eq
Dairy ⁶	2 cups	2½ cups	2½ cups	3 cups	3 cups	3 cups	3 cups	3 cups	3 cups	3 cups	3 cups	3 cups
Oils ⁷	15 g	17 g	17 g	22 g	24 g	27 g	29 g	31 g	34 g	36 g	44 g	51 g
Limit on calories from SoFAS ⁸	137	121	121	121	161	258	266	330	362	395	459	596

4. THE FIVE FOOD GROUPS

a. PURPOSE OF FOOD GROUPS

- *Food Group Benefits:* Simplify dietary recommendations by focusing on foods instead of nutrients.
- *USDA Food Patterns:* Provide recommended amounts of each food group and subgroup at 12 different calorie levels.
- *Nutritional Needs:* Meeting nutritional needs without tracking individual nutrients.

b. FOOD GROUP AMOUNTS

- *USDA Food Patterns Targets:* Specify targets for each food group in cup or ounce equivalents.
- *Calorie Allowance:* Each pattern includes a limited number of calories for added sugars and saturated fats.
- *Nutrient-Dense Food Choices:* Americans are encouraged to choose nutrient-dense foods and limit added sugars and saturated fat to less than 10% of total calorie intake.

1.2 - 2. CALCULATING NUTRIENT NEEDS

Q. A client you are working with weighs 56kg. How many pounds is that?
A. 56*2.2=123.2lbs

Q. A male that rides their bike 30 minutes each day would get which physical activity factor? 1.11

Q. A female that has a sedentary job who does not engage in any physical activity would have which physical activity factor? 1

1. THE IMPORTANCE OF NUTRIENTS

- *Nutrient Importance:* Nutrient needs vary based on individual factors like lifestyle, age, height, and activity level.
- *Macronutrients:* Carbohydrates, fats, and proteins are essential macronutrients that provide energy and contribute to overall well-being.
- *Caloric Needs:* Caloric intake should be based on individual energy requirements, which can be estimated using formulas like BMR or dietary reference intake.

2. CALCULATING YOUR ESTIMATED ENERGY REQUIREMENT

- To determine your **estimated energy requirement (EER)**, use the appropriate equation (in equations below), inserting your age in years, weight (wt) in kilograms, height (ht) in meters, and physical activity (PA) factor from the accompanying table.
- *EXAMPLE:*
 - Consider an active 25-year-old male who is 5 feet 10 inches tall and weighs 170 pounds. First, you convert his weight from pounds to kilograms and his height from inches to meters, if necessary:
 - Weight: $170/2.2$
 - Height: $70'' = 70/39.37$
 - Consider his level of daily physical activity and select the appropriate PA factor from the table. Then, insert his age, PA factor, weight, and height into the appropriate equation:
 - $662-(9.53*A)+PA*[(15.91*W)+539.6*H]$
 - $662-(9.53*25)+1.25*[(15.91*(170/2.2))+539.6*(70/39.37)] = 3159.77$

1.2 - 3. RESOURCES TO PLANNING A HEALTHY DIET

Q. Which of the following is true regarding the RDA (Recommended Daily Allowance)? It is the average daily dietary intake level that is sufficient to meet the nutrient requirement of nearly all healthy individuals in a group.

Q. The bag of pretzels you ate has 10g of carbohydrates. How many calories from carbs would you have eaten in those 10g? 40

1. PLANNING A DIET

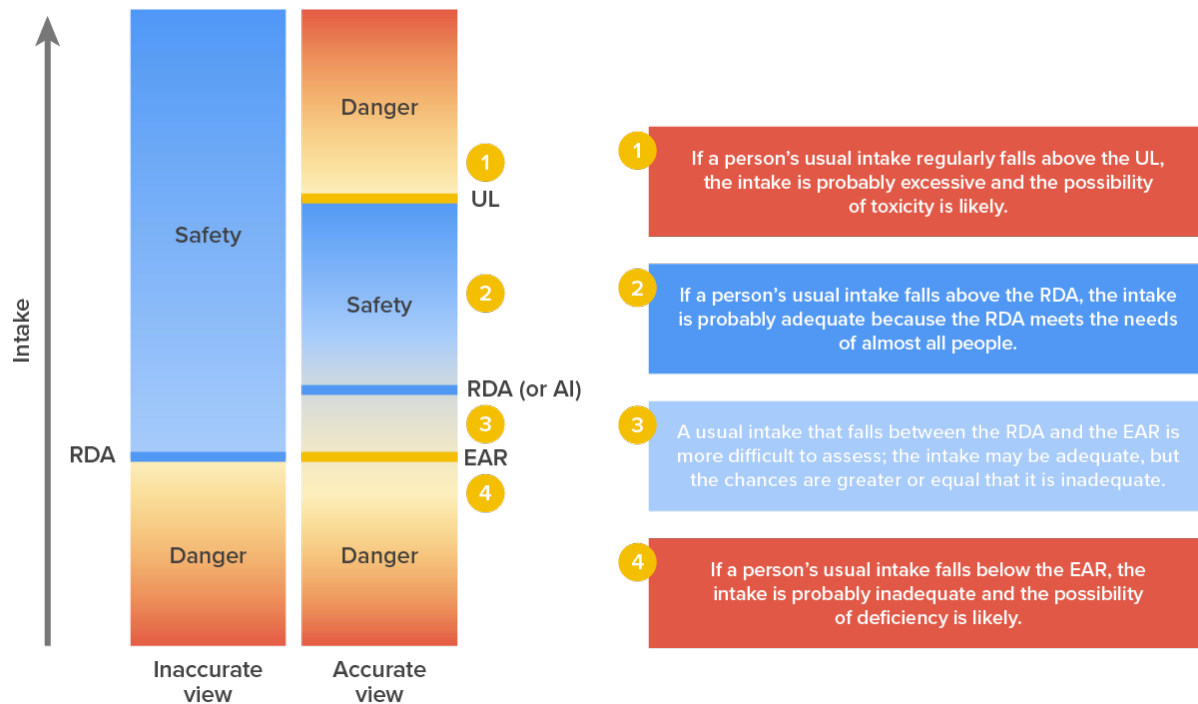
- A diet is anything that is consumed by a person or people on a regular basis.

2. TOOLS FOR DESIGNING HEALTHY DIETS

- *Acceptable Macronutrient Distribution Range (AMDR):* The AMDR describes the proportions of daily caloric intake that should be carbohydrates, lipids (fat), and proteins. Basically, the AMDR provides guidelines on how many macronutrient calories one should consume a day.
- According to the AMDR, the range of caloric intake in a daily diet should be:
 - Carbohydrates: 45-65%

- Lipids (fat): 20-35%
- Proteins: 10-35%
- There are 4 calories per gram for carbohydrates, 4 calories per gram for protein, and 9 calories per gram for fat.

3. DIETARY REFERENCE INTAKES (DRI)



4. SELECTING FOODS TO BUY

- Minimally processed

1.2 - 4. TOOLS FOR PLANNING A HEALTHY DIET

Q. Your breakfast has 10g of fat and 10g of protein. How many total calories?
A. $10 \times 9 + 10 \times 4 = 130$

Q. A large cheeseburger gives you 30g of fat, 30g of carbs, 15g of protein. Calories?
A. $30 \times 9 + 30 \times 4 + 15 \times 4 = 450$

Q. If Mary eats a bowl of 18 chips, with 9 chips per serving, how many grams of fat will she consume (based on label with 5g per serving)? $(18/9) \times 5 = 10$

1. WHAT ARE CALORIES AND MACRONUTRIENTS?

- *Calorie Source:* Calories, the energy in food, come from carbohydrates, proteins, and fats.
- *Macronutrient Energy:* Carbohydrates and proteins provide 4 calories per gram, while fats provide 9 calories per gram.

- *Macronutrient Examples:* Carbohydrates include bread, pasta, and fruits; proteins include meat, dairy, and beans; fats include oils, nuts, and avocados.
- *Carbohydrates:* provide the most energy to the body and are the primary source of energy for the brain.
- *Proteins:* building blocks for the structure and function of cells. Protein sources are important to provide energy and rebuild muscles.
- *Saturated fat:* found mostly in animal-based foods, such as beef and full-fat dairy.
- *Unsaturated fat:* found in most plant-based foods, such as olive oil, peanut oil, avocados, and peanut butter.
- *Trans fat:* found in foods such as pizza, doughnuts, and cookies. Trans fats are typically not natural, but instead, are chemically produced to extend the shelf life of food.

VOCAB

- *Dietary Guidelines for Americans (DGA):* A set of guidelines that are updated every five years to guide policymakers, nutrition experts (like Registered Dietitian Nutritionists), and other health professionals to promote health and prevent disease.
- *Dietary Pattern:* The combination of foods and beverages that constitutes an individual's complete dietary intake over time. This may be a description of a customary way of eating or a description of a combination of foods recommended for consumption.
- *Food Group Plan:* A plan to help you build a diet from clusters of foods that are similar in nutrient content.
- *The Five Food Groups:* Fruits, Vegetables, Grains, Protein Foods, and Dairy.
- *United States Department of Agriculture (USDA):* A department responsible for developing and executing laws related to farming, forestry, rural economic development, and food.
- *Basal Metabolic Rate (BMR):* The rate at which the body uses energy while at rest to keep vital functions going, such as breathing and keeping warm.
- *Estimated Energy Requirement (EER):* The average dietary energy intake that is predicted to maintain energy balance in a healthy adult of a defined age, gender, weight, height, and a level of physical activity that is consistent with good health.
- *Acceptable Macronutrient Distribution Range (AMDR):* Provides guidelines on how many macronutrient calories one should consume a day.
- *Diet:* Anything that is consumed by a particular person or people on a regular basis
- *Dietary Reference Intakes or DRIs:* are reference values of nutrient intake that help with nutrition planning and assessment of healthy individuals.
 - *Recommended Dietary Allowance (RDA):* the average daily dietary intake level that is sufficient to meet the nutrient requirement of nearly all (about 97%) healthy individuals in a group. This is the basic quantity of a nutrient recommended.
 - *Adequate Intake (AI):* a value based on observed or experimentally determined approximations of nutrient intake by a group (or groups) of healthy people—used

when an RDA cannot be determined. This is the minimum amount of a nutrient needed for maintaining health.

- *Tolerable Upper Intake Level (UL)*: the highest level of daily nutrient intake that is likely to pose no risk of adverse health effects to almost all individuals in the general population. As intake increases above the UL, the risk of adverse effects increases. This is the maximum that would be consumed prior to developing negative effects of eating too much. This is not a level that is met but rather one that is avoided to prevent a decrease in health.
- *Estimated Average Requirement (EAR)*: a nutrient intake value that is estimated to meet the requirement of half the healthy individuals in a group. These nutrient values should be used as goals for dietary intake for health.

EQUATIONS

• *Estimated Energy Requirement*

- For men 19+: $EER = 662 - (9.53 \cdot A) + PA \cdot [(15.91 \cdot W) + 539.6 \cdot H]$
- For women 19+: $EER = 354 - (6.91 \cdot A) + PA \cdot [(9.36 \cdot W) + 726 \cdot H]$
- *Pounds to Kg: divide by 2.2; Kg to Pounds: multiply by 2.2.*
- *Inches to meters, divide by 39.37; Meters to inches, multiply by 39.37.*

EER = estimated energy requirement;

A = age, in years;

W = weight, in kilograms;

H = height, in meters; and

PA = physical activity, with the following values:

PA	Males	Females
Sedentary Typical daily living activities	1.0	1.0
Low active plus 30–60 min moderate activity	1.11	1.12
Active plus ≥ 60 min moderate activity	1.25	1.27
Very active plus ≥ 60 min moderate activity and 60 min vigorous or 120 min moderate activity	1.48	1.45